

Experiment7

- 1) Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Code :

```
#include <stdio.h>
#include <stdlib.h>

int global_var = 10; // Global variable

void code_function()
{   printf("Code address  : %p\n", (void
*)code_function);
}

int main() {   static int static_var = 20; //
Static variable   int stack_var = 30;      //
Stack variable
    int *heap_var = malloc(sizeof(int)); // Heap variable

    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
return 1;
    }

    *heap_var = 40;

    printf("Linux Process Address Space\n");   printf("----
-----\n");

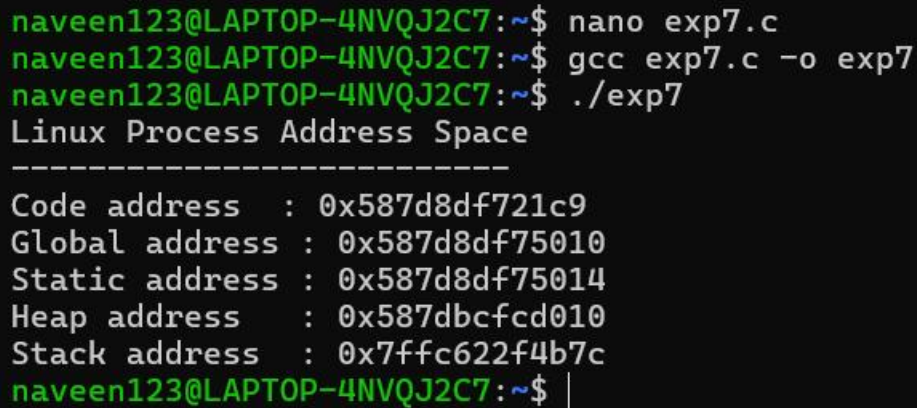
    code_function();

    printf("Global address : %p\n", (void *)&global_var);
    printf("Static address : %p\n", (void *)&static_var);
    printf("Heap address   : %p\n", (void *)heap_var);
    printf("Stack address  : %p\n", (void *)&stack_var);
```

```
    free(heap_var);

    return 0;
}
```

OUTPUT :



```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp7.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp7.c -o exp7
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp7
Linux Process Address Space
-----
Code address   : 0x587d8df721c9
Global address : 0x587d8df75010
Static address : 0x587d8df75014
Heap address   : 0x587dbcfcd010
Stack address  : 0x7ffc622f4b7c
naveen123@LAPTOP-4NVQJ2C7:~$ |
```

2) Make use of the /proc//maps file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings.

Code:

```
#include <stdio.h>

#include <stdlib.h>

#include <unistd.h>


int global_var = 100;


int main()
{
```

```
static int static_var = 200;

int stack_var = 300;

int *heap_var = malloc(sizeof(int));

*heap_var = 400;

printf("Process ID (PID): %d\n", getpid());

printf("Global variable address : %p\n", (void *)&global_var);

printf("Static variable address : %p\n", (void *)&static_var);

printf("Heap variable address : %p\n", (void *)heap_var);

printf("Stack variable address : %p\n", (void *)&stack_var);

printf("\nProcess is running...\n");   getchar();

free(heap_var); return 0;

}
```

OUTPUT:

```
naveen123@LAPTOP-4NVQJ2C7:~$ nano exp71.c
naveen123@LAPTOP-4NVQJ2C7:~$ gcc exp71.c -o exp71
naveen123@LAPTOP-4NVQJ2C7:~$ ./exp71
Process ID (PID): 661
Global variable address : 0x64ae6c33d010
Static variable address : 0x64ae6c33d014
Heap variable address   : 0x64aea6c3e010
Stack variable address  : 0x7fff6817bbbc

Process is running...
|
```